

Provenance for every number in main.tex

Rule for this paper: **no number goes in the draft unless it traces to a job log or a file in this repo**. This project has already been burned twice by figures that turned out to be the wrong column or an agent’s unverified claim, so the audit trail is part of the deliverable.

Status legend: **[V]** I verified this myself · **[A]** from an agent report, not independently re-checked · **[P]** published, quoted from a paper.

Section 3 — the two-microphone measurement (Table 1)

Source: results/amt_bridge_eval-437566.log, produced by scripts/amt_bridge_eval.py via amt_bridge_eval_cpu.sh. Transcriptions from job 402243 (scripts/amt_transcribe_real.py), stored under /scratch/pmohseni/amt_out/{kong_stock,edwards_robust}/{room,di-left}/ — 25 pages per cell, 100 files total.

cell	value	status
kong room F1@50ms	0.9116	[V]
kong di-left F1@50ms	0.9124	[V]
kong room F1@100ms	0.9546	[V]
kong di-left F1@100ms	0.9903	[V]
edwards room F1@50ms	0.9441	[V]
edwards di-left F1@50ms	0.9259	[V]
edwards room F1@100ms	0.9864	[V]
edwards di-left F1@100ms	0.9909	[V]

Round-trip validation, same log: - oracle at latency 0: **100.00** pct@0.5s **[V]** — proves the score-note → pixel → onset-frame path is lossless. - oracle at latency −0.050: **94.79** **[V]** — independently reproduces an earlier in-repo figure of 93.4 under the same convention.

Non-causal / online figures: - offline DTW, room: **98.06** **[V]** - online greedy, room: **10.73**; di-left **26.69**; median error 8-13 s **[V]**

Measured systematic offset: constant wav-vs-MIDI shift of **−0.03 to −0.04 s** across all four conditions **[V]** — under one frame at 20 fps.

Caveat that must stay in the paper: score/{piece}.mid is byte-identical to performance/{piece}.mid — these are reproducing-piano playback, so there is **no tempo deviation**. This is why the DTW figure is near-tautological and must not be presented as a score-following result.

Section 4 — the intervention asymmetry

claim	value	status
detection tracker, no IR → IR, room	46.0 → 71.2 (+25.2)	[P] Henkel & Widmer, EUSIPCO 2021
our heatmap tracker, fixed real IR	45.6 → 56.6 (+11.0)	[V]
IR bank size	693 real measured IRs (/scratch/pmohseni/ir_bank)	[V]

claim	value	status
CYOLO loads after its own filter	663	[V] results/cyolo_repro-547365.log

Our 56.6: results/eval_any-497726.log (R2r_realir round 2) **[V]**. Round 1 = 56.2, round 3 = 54.4 — **the run peaked at round 2 and then regressed**, so 56.6 is final, not a waypoint. Companion tiers: di-left 64.5, synth 87.4 **[V]**.

The bug disclosure: -extensions/augmentation/impulse_response.py used `fftconvolve(mode='same')`, advancing audio by $(\text{len}(\text{ir})-1)/2$ samples — a **4.1-20.0 frame** label desync on ~50% of samples. Fixed in commit e8320ea, **2026-08-06**. **[V]** - The discarded IR experiment (B6) trained **2026-07-05 - 07-08**, a month before the fix. **[V]** So “IR augmentation does not help us” was reading a bug. - CYOLO’s own convolution is `convolve(..., 'full')[:-(len(ir)-1)]`, which is mathematically identical to our *fixed* version **[V]** — good evidence the fix is right rather than merely different.

Output-parameterisation comparison (same paper, same data, same audio tower): MM-Loc **58.5** vs CUNet **22.4** on room **[P]**, Frontiers Table 4.

Section 5 — measurement problems

Aggregation — AGGREGATION_FINDING.md, recomputed from results/eval_any-111639.log (25 pages, 4415 onsets):

estimator	value	status
micro (pooled onsets)	45.4	[V]
macro over 25 pages	46.8	[V]
macro over 16 pieces	52.7	[V]

Chopin Op.9 = 29.8% of the micro metric, 6.2% of piece-macro; scores 14.6 where the rest scores 57.0 **[V]**.

Power — scripts/benchmark_power_analysis.py, results/analysis/benchmark_power.json: design effect ≈ 180 , effective $N \approx 25$ pieces, MDE ≈ 10 points **[V]**. Retrain control: same recipe, two runs $\rightarrow 22.7$ vs 16.4 **[V]**.

Oracle inputs — CODA_COMPARABILITY.md, audited at commit dba9829: `system_boxes/bar_boxes` are required positional args of `forward()` (`coda_model.py:314-317`), passed at the training call site (`train.py:173-177`); loss has only system CE + bar CE + note MSE, no localisation term **[V]**. No detection path exists in the release **[V]**.

Baseline figures

system	room @0.5s	status
cyolo_sb (reproduced by us)	79.9	[V] --only_onsets, matches published row exactly
cyolo_sb @0.1s	63.0	[V]
CYOLO	71.2	[P]
cyolo_sb_a (not reproducible — needs “+A” data)	86.5	[P]

system	room @0.5s	status
CUNet	22.4	[P]
MM-Loc	58.5	[P]

The 17-point column error (63.0/70.6 are the **0.1 s** column, not 0.5 s) is documented in BASELINE_CORRECTION.md **[V]**.

Deliberately NOT in the paper

- **P1 (bucketed-softmax output head).** Three failed runs, three different causes (module-level import; centre-of-mass over an above-half-max plateau; height marginalisation on multi-staff pages). It has **never been measured** — neither supported nor refuted. Nothing about it goes in until a clean number exists.
- **OMR results** (oemer F1 0.985 native / 0.9925 hires, median $|\Delta x|$ 0.33 px). Verified **[V]**, but off-thesis for a 4-page paper. Belongs in the longer version.
- **The full CODA audit.** Too big to defend in 4 pages; Section 5 states only the part that bears on measurement. Full version → TISMIR/ISMIR.
- **cyolo+IR reproduction.** Still training (epoch ~10/50). If it converges near 71.2 it strengthens Section 4 considerably; if it undershoots, Section 4's asymmetry claim weakens and must be softened.

Open risks to the draft

1. Section 4 leans on one published delta (+25.2) against one measured delta (+11.0). Our own cyolo+IR run is the control that makes this a like-for-like comparison; until it converges, the comparison is across papers.
2. The two-mic result is on reproducing-piano playback. A human performance would stress transcription harder. This bounds the claim and is stated.
3. No ICASSP precedent found for an evaluation-audit paper, which is why the thesis is the positive measurement (Section 3) and not the audit.